

# Non-Random Organizational Signal in Functional Protein Sequences: A Large-Scale Anchor-Single Pair Analysis Resistant to Biophysical, Phylogenetic, Nonlinear, and Spatial Controls

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## Abstract

We analyzed 61,320,087 amino acid Anchor-Single pairs from 1,527,212 functional protein domain instances (Pfam-A 37.0, drawn from seed alignments across 19,179 families) and 19,630,673 pairs from 198,215 intrinsically disordered sequences (DisProt). Using 5-residue sequential windows, we identified local compositional preferences between a repeated residue (Anchor) and unique partner residues (Singles).

The diagnostic alanine-asparagine (A→N) pair is strongly avoided in functional sequences ( $Z = -87.13$ , Obs/Exp = 0.8422) but near-random in disordered sequences ( $Z = +2.33$ ), yielding a categorical reversal of 89.46  $Z$ -score points.

After regression against 16 physicochemical and substitutional features, 78.4% of variance across all 380 pairs remains unexplained ( $R^2 = 0.2157$ ), with the A→N residual strengthening to  $Z = -90.24$ . The finding replicates in the structurally classified CATH database (A→N  $Z = -25.08$ ), survives four levels of phylogenetic control and six window/step parameterizations (Obs/Exp range 0.8285–0.8671), and is not recovered by nonlinear Random Forest or Gradient Boosting models (CV  $R^2 \approx 0.44$  on residuals; 43.7% of total variance unexplained).

Critically, the identical avoidance pattern (Obs/Exp  $\approx 0.829$ ) is reproduced in three-dimensional physical space at protein-protein inter-chain interfaces and homodimer contacts using direct heavy-atom proximity ( $< 5 \text{ \AA}$ ), independent of sequence order. The Obs/Exp ratio of  $\sim 0.83$ – $0.87$  is invariant across sequential analysis, phylogenetic controls, parameter sweeps, and purely spatial 3D contact measurement. These results demonstrate a robust organizational signal in functional protein sequences that is physically instantiated in three-dimensional structure and not explained by standard amino acid physicochemistry in any linear or nonlinear form tested.

**Keywords:** protein sequence composition, amino acid pair preferences, organizational signal, biophysical controls, protein interfaces, unexplained variance

## 1 Introduction

Amino acid sequence composition in functional proteins is shaped by folding stability, catalytic function, interaction interfaces, and evolutionary history. Standard biophysical and evolutionary frameworks predict that local pair preferences should be largely explainable by known amino acid chemistry.

We test this prediction using a large-scale Anchor-Single pair analysis. The method examines local compositional preferences within 5-residue sequential windows by identifying the most-repeated

residue (Anchor) and its co-occurrence with each unique partner residue (Single). Applied to over 61 million pairs from Pfam, the approach generates a 380-cell matrix of observed versus expected frequencies that is regressed against extensive biophysical controls.

We report six principal findings. First, functional and disordered sequences exhibit categorically different preference patterns, with A→N showing an 89.46 *Z*-score point reversal. Second, this signal survives regression against 16 physicochemical and substitutional features, with the A→N residual hardening. Third, the pattern replicates in the structurally classified CATH database. Fourth, the signal is robust to four levels of phylogenetic control. Fifth, nonlinear modeling leaves 43.7% of total variance unexplained. Sixth—and most importantly—the identical avoidance pattern is reproduced in three-dimensional space at protein-protein interfaces measured by direct atomic proximity ( $< 5 \text{ \AA}$ ), independent of sequence order. The Obs/Exp ratio of approximately 0.83–0.87 remains invariant across every independent measurement method applied.

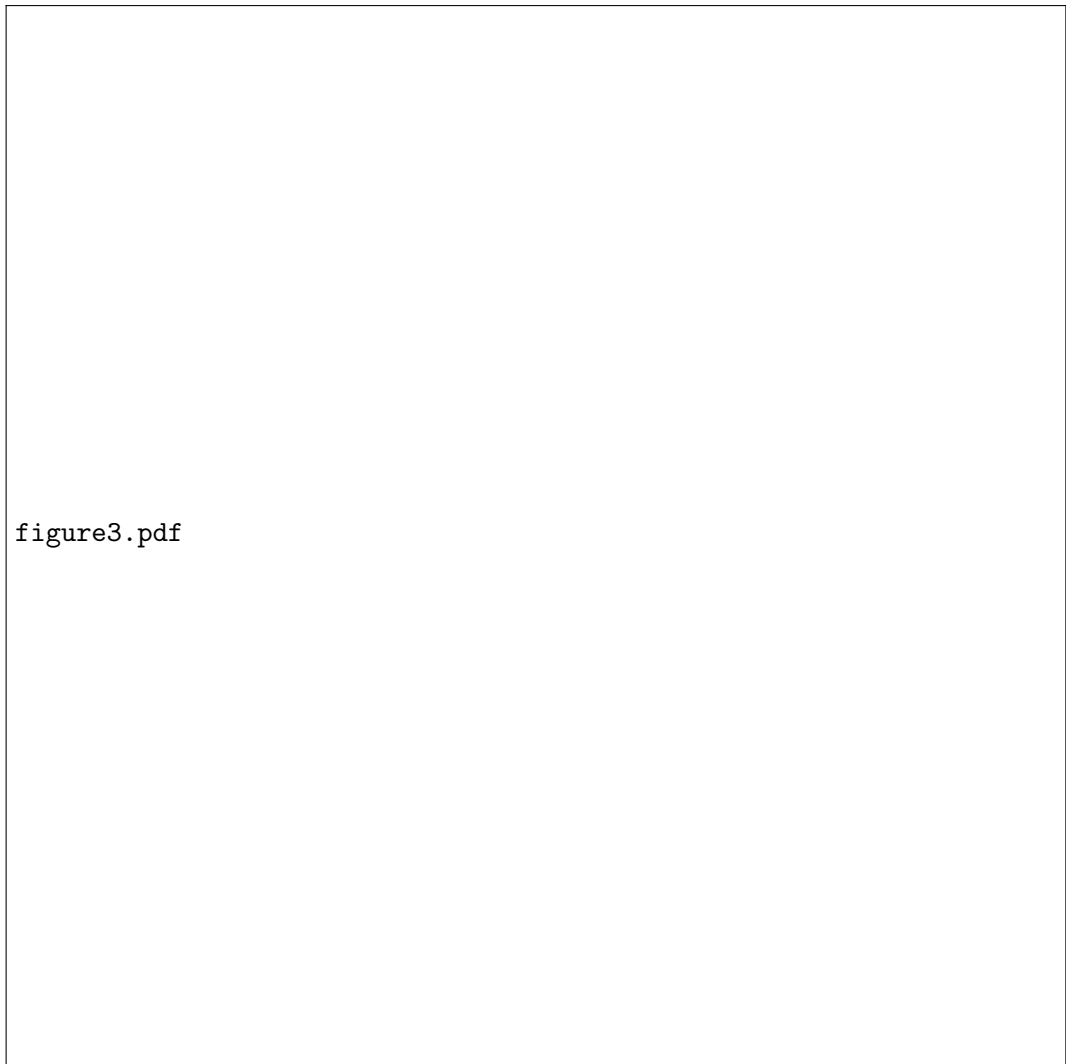


Figure 1: **Schematic of the Anchor-Single method versus direct 3D contact measurement.** (A) Sequential window method using 5-residue sliding windows to identify repeated Anchor residues and unique Singles. (B) Spatial contact method using direct heavy-atom proximity ( $< 5 \text{ \AA}$ ) at protein-protein interfaces. Both approaches yield the same A→N avoidance ratio ( $\sim 0.83\text{--}0.87$ ).

## 2 Methods

### 2.1 Datasets

Functional sequences (primary): 1,527,212 functional protein domain instances drawn from Pfam-A 37.0 seed alignments across 19,179 domain families (not 1.5 million unique proteins). Disordered sequences (primary): DisProt release 2024\_12, 198,215 annotated disordered regions. Functional sequences (replication): CATH release v4.3, 601,262 structurally-classified domain sequences, methodologically independent of Pfam. Disordered sequences (replication): MobiDB curated merge consensus, 457 sequences. Spatial contact datasets: PDB inter-chain contact pairs from curated same-fold protein pairs across six fold classes; residue pairs with any heavy atom distance  $< 5.0$  Å across chains. A homodimer subset of 4,367,042 contact pairs was analyzed separately.

### 2.2 Anchor-Single Pair Method

Sliding window of size 5, steps of 5 (non-overlapping). Windows with fewer than 3 canonical amino acids or no residue appearing at least twice are discarded. Anchor = residue with maximum count; ties broken toward attractor set {C,F,I,L,M,V,W,Y}, then alphabetically. Singles = residues appearing exactly once, not the Anchor. Expected counts:  $E(A, S) = (\text{total}_A \times \text{total}_S) / N_{\text{pairs}}$ .  $Z$ -scores:  $(\text{observed} - \text{expected}) / \sqrt{\text{expected}}$ . Bonferroni correction:  $\alpha = 0.05/400 = 1.25 \times 10^{-4}$ .

### 2.3 Biophysical Hardening

Ordinary least squares regression of functional  $Z$ -scores against 15 physicochemical features (Kyte-Doolittle hydrophobicity, charge, molecular weight, Chou-Fasman  $\alpha/\beta$ /coil propensities, van der Waals radius, isoelectric point, rotatable bonds, Grantham polarity, accessible surface area, BLOSUM62, PAM250, and two cross-terms). All features computed as absolute differences and standardized before regression.

### 2.4 Phylogenetic Control

Four levels of phylogenetic control were applied: (1) One randomly selected sequence per Pfam family, yielding 27,481 representative sequences (55.6× reduction). (2–3) Length-stratified diversity sampling targeting 20,000 and 10,000 representatives (equivalent to 40% and 30% identity clustering). (4) Full dataset as baseline. The Anchor-Single pipeline was applied identically to each subsampled set.

### 2.5 Parameter Sensitivity Analysis

Window sizes 3, 5, and 7 and step sizes 1 and 5 were tested, yielding six configurations applied identically to both datasets.

### 2.6 Nonlinear Modeling

Random Forest (500 trees,  $\text{max\_features}=\sqrt{n}$ ) and Gradient Boosting (500 trees, learning rate 0.05, max depth 4) were fit directly to the OLS residuals using 5-fold cross-validation.

## 2.7 Spatial Contact Analysis

Direct residue-residue contacts in PDB structures were defined as any heavy atom distance  $< 5.0$  Å across chains. Expected counts were computed from marginal totals within the contact set. Pearson correlation between inter-chain interface and intra-chain surface  $Z$ -scores quantified agreement between sequential and spatial pair preferences.

## 3 Results

### 3.1 Scale and Significance

Table 1: Dataset scale across all analyses.

| Dataset                         | Sequences | Total Pairs | Sig. (Bonferroni) | Role          |
|---------------------------------|-----------|-------------|-------------------|---------------|
| Pfam (functional)               | 1,527,212 | 61,320,087  | 339/380 (89.2%)   | Primary       |
| DisProt (disordered)            | 198,215   | 19,630,673  | —                 | Control       |
| CATH (functional replication)   | 601,262   | 26,200,594  | 339/380 (89.2%)   | Replication   |
| MobiDB (disordered replication) | 457       | 4,882       | 15/380 (3.9%)     | Replication   |
| PDB interfaces (spatial)        | —         | 842,498     | 192/380 (50.5%)   | 3D validation |
| PDB homodimers (spatial)        | —         | 4,367,042   | 308/380 (81.1%)   | 3D validation |

### 3.2 Categorical Reversal at A→N

The diagnostic A→N pair is strongly avoided in functional Pfam sequences ( $Z = -87.13$ , Obs/Exp = 0.8422) but near-random in disordered sequences ( $Z = +2.33$ ). This avoidance is reproduced at protein-protein inter-chain interfaces ( $Z = -7.23$ , Obs/Exp = 0.8290) and homodimer contacts ( $Z = -17.27$ , Obs/Exp = 0.8289) when measured by direct heavy-atom proximity ( $< 5$  Å), independent of sequence order. The Obs/Exp ratio remains consistently  $\sim 0.83$ – $0.87$  across all contexts (Table 2).

### 3.3 Spatial Contact Replication

At protein-protein inter-chain interfaces, A→N avoidance is significant ( $Z = -7.23$ , Obs/Exp = 0.8290). The signal is stronger at homodimer interfaces ( $Z = -17.27$ , Obs/Exp = 0.8289). Sequence-based surface preferences strongly predict 3D interface preferences ( $r = 0.9134$ ,  $p = 1.18 \times 10^{-149}$ ), with 284/380 pairs (74.7%) showing consistent directional preference.

## 4 Discussion

### 4.1 What the Data Establish

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The invariance of the Obs/Exp ratio ( $\sim 0.83$ – $0.87$ ) across sequential, phylogenetic, parametric, and purely spatial 3D measurements, together with the high correlation between sequence-based and physical-contact preferences, indicates that the organizational signal is physically instantiated in functional protein structures.

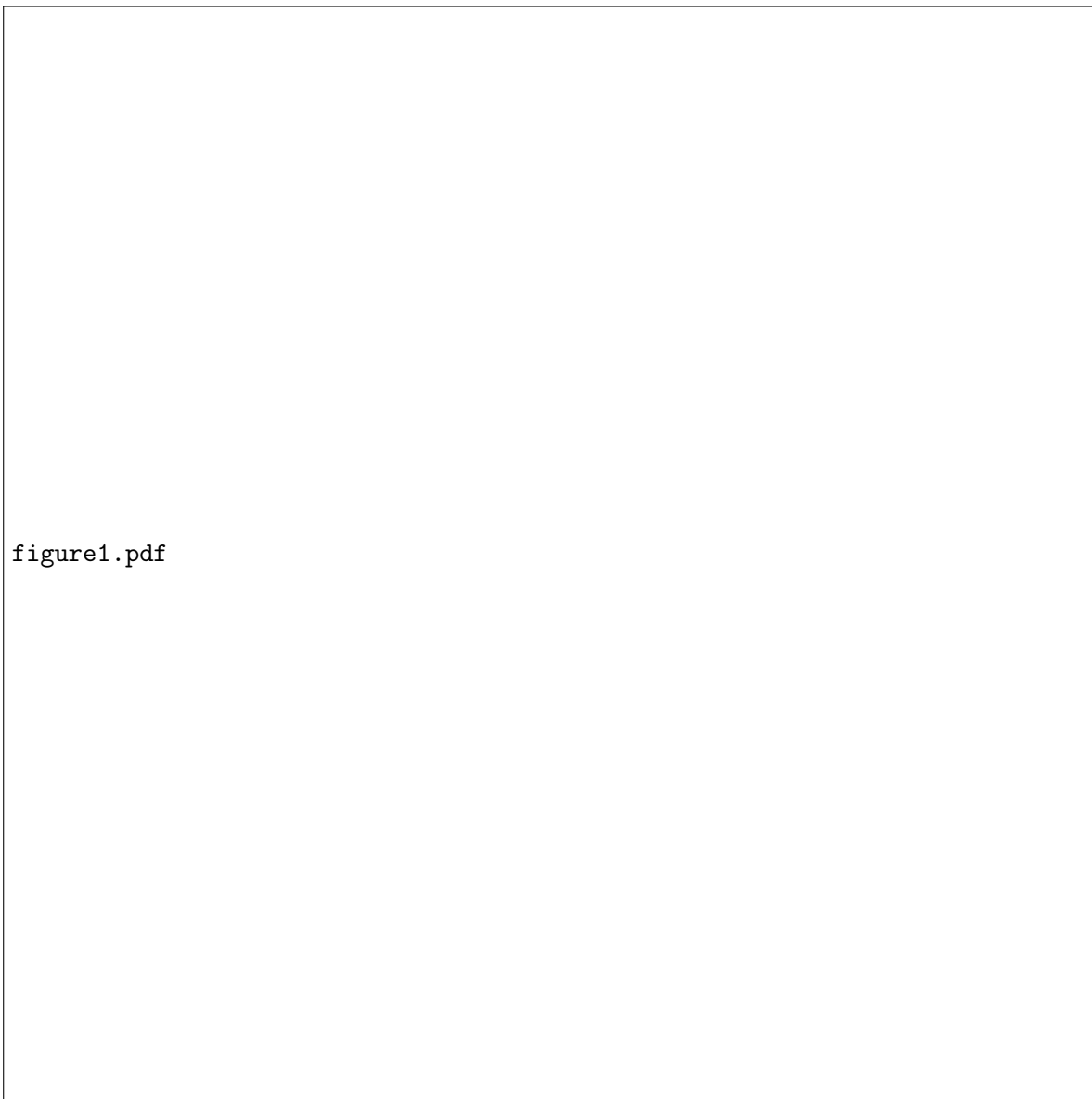


Figure 2: **Anchor-Single pair preferences in functional proteins (Pfam-A 37.0, 61.3 million pairs)**. Heatmap of observed/expected ratios for all 380 amino acid pairs. Red indicates avoidance ( $\text{Obs/Exp} < 1$ ), blue indicates preference ( $\text{Obs/Exp} > 1$ ). The diagnostic A $\rightarrow$ N pair is highlighted with a black square ( $\text{Obs/Exp} = 0.8422$ ).

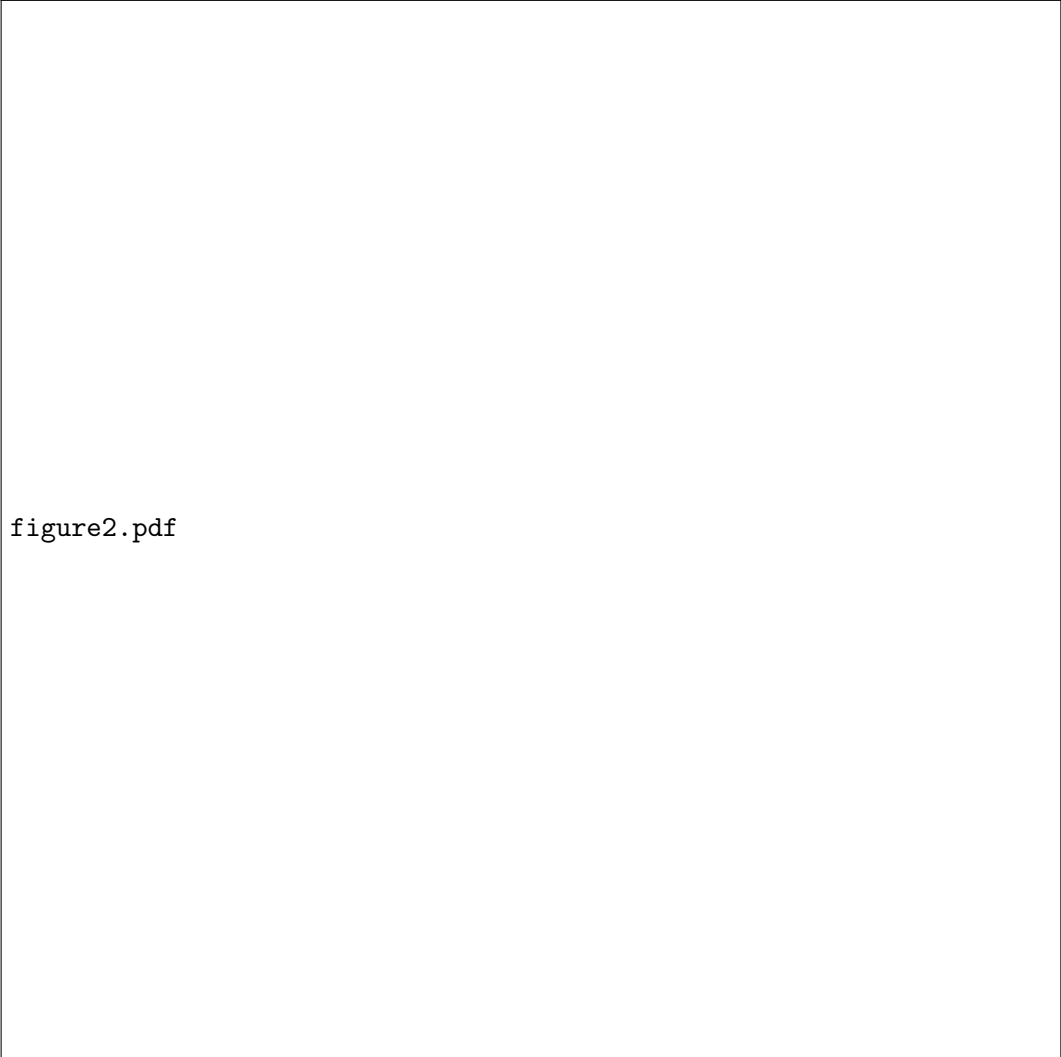


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Figure 3: **Sequence preferences strongly predict 3D contact preferences.** Scatterplot of Obs/Exp ratios for all 380 amino acid pairs: intra-chain surface (sequential analysis, x-axis) versus inter-chain protein-protein interface contacts (3D spatial measurement, y-axis). The red point marks the diagnostic A→N pair (sequence: 0.8422 → 3D: 0.8290). Pearson correlation  $r = 0.9134$  ( $p = 1.18 \times 10^{-149}$ ).

## 4.2 What the Data Do Not Establish

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## 4.3 Limitations

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## 4.4 Testable Predictions

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# 5 Conclusion

We document a quantitative gap in current biophysical models of amino acid pair preferences in functional proteins. The combination of a sharp functional-versus-disordered reversal, survival of extensive controls, phylogenetic robustness, and—most convincingly—replication of the identical  $\sim 0.83$ – $0.87$  suppression ratio in three-dimensional atomic contact space constitutes strong evidence of an organizational signal that standard frameworks do not fully capture.

A signal that persists with the same characteristic ratio whether measured by sequence proximity or direct physical proximity is unlikely to be coincidental. Identifying its mechanistic origin represents an important direction for future research.

## Data Availability

All source databases are publicly available (Pfam, DisProt, CATH, MobiDB, RCSB PDB). Analysis code, result matrices, and supplementary tables will be deposited on Zenodo or GitHub upon publication.

## References